1.4.2 Reducing costs through predictive maintenance

There is a good reason that digital twins are often found today in industries that operate complex machinery. In such operations, the failure of a machine that runs continuously can have major consequences financially and environmentally, in terms of health and safety, or regulatory repercussions. Being able to gain the insight that a machine is starting to show anomalous behavior allows the operator to examine and repair it before it fails, thereby minimizing the risk of downtime in an approach known as *predictive maintenance*.

REAL WORLD EXAMPLE: ROLLS ROYCE INTELLIGENT ENGINE

You may have experienced the frustration of a flight delay where you are stuck in a terminal with no information as to what has happened and when you may be on your way again. Unplanned maintenance is one of the leading causes of flight delays so being able to predict when an engine might require unscheduled maintenance benefits both airlines and their customers. Rolls-Royce created a program known as the "Blue Data Thread" that provides bidirectional flow of information—a digital representation of the physical jet engine—between the company and the airlines operating their engines. These digital twins of aircraft engines, known as the "Intelligent Engine", are used to predict what maintenance will be required on which engines at what time, and ensure that repair capabilities and capacity will match these requirements. When engines have been sold for many years under long term maintenance contracts and the cost of engine maintenance is transferred back to the company, they are incentivized to optimize this process as much as possible.

1.4.3 Optimizing performance and operational efficiency

The increased visibility into the operations of plant and equipment that having a digital replica brings, together with the ability to predict and simulate the impact of changes, is a key factor in being able to improve operations across many different industries.